Royal Crest Academy

SCIENCE · INNOVATION · SKILL

FREE LEARNER GUIDE

Scientific method & lab safety guide

Biology, Chemistry, Physics and Mathematics taught by specialists, with an emphasis on practical work, clear explanation and disciplined exam technique.

The scientific method

1 Observe
Notice something interesting in the world around you.

2 Question
Ask a clear question that can be tested.

3 Hypothesise
Predict what will happen, and say why.

4 Experiment
Test it fairly: change one variable and keep the others the same.

5 Analyse
Record results in a table, draw a graph and look for patterns.

6 Conclude
Answer the question, explain the result and share it.

Variables in a fair test

Independent variable	The one thing you change on purpose.
Dependent variable	The thing you measure to see the effect.
Control variables	Everything you keep the same, so the test is fair.
Reliable results	Results that are similar when the experiment is repeated.
Anomaly	A result that does not fit the pattern of the others.
Hypothesis	A testable prediction with a scientific reason.

Lab safety: the golden rules

1. Never enter the lab without a teacher.
2. Wear goggles and a lab coat for practical work.
3. Tie back long hair and tuck in ties.
4. Never eat, drink or taste anything in the lab.
5. Report every breakage, spill or injury at once.
6. Read labels and hazard symbols before using a chemical.
7. Point test tubes away from people when heating.
8. Wash your hands when you finish.

Plan your own investigation

MY QUESTION

MY HYPOTHESIS (AND WHY)

INDEPENDENT VARIABLE

DEPENDENT VARIABLE

CONTROL VARIABLES

EQUIPMENT

RESULTS TABLE

CONCLUSION

Check yourself

1. In an investigation into how temperature affects how fast sugar dissolves, what is the independent variable?

2. Why do scientists repeat experiments?

3. What must you do first if you spill a chemical?

Answers 1. Temperature: it is the variable you change on purpose. · 2. To check that the results are reliable and to spot any anomalies. · 3. Tell the teacher straight away.